PROJECT PROPOSAL

Title: Hotel Booking Data Analysis

1. Student Details

- **Name:** Pranita Vinodkumar Patil
- **Email:** pranitapatil912004@gmail.com
- **Class/Division:** TY. ECM
- **Roll Number:** 66

---

2. Problem Statement

In the hospitality industry, hotels face several challenges such as high booking cancellations, seasonal demand fluctuations, and changing customer preferences. These issues can affect revenue and operational efficiency.

The aim of this project is to analyze hotel booking data to understand customer behavior, identify booking trends, and find key factors that influence cancellations. This analysis will help in making better business decisions.

---

3. Domain

Hospitality / Tourism / E-commerce

---

4. Dataset Description

- **Source:** Kaggle
- **Dataset Name:** Hotel Booking Demand Dataset
- **Link:** [https://www.kaggle.com/datasets/jessemostipak/hotel-booking-demand] (https://www.kaggle.com/datasets/jessemostipak/hotel-booking-demand)

Dataset Features:

The dataset contains the following information:

- Type of hotel (City Hotel or Resort Hotel)
- Booking dates (year, month, day)
- Length of stay
- Number of adults, children, and babies
- Booking cancellation status
- Average Daily Rate (ADR)
- Customer details and booking information

This dataset is useful for analyzing booking patterns and customer behavior.

---

5. Objectives

- To analyze hotel booking trends
- To identify factors affecting booking cancellations
- To study seasonal variations in bookings
- To visualize data using graphs and charts
- To generate useful insights for decision-making

---

6. Methodology

The project will follow these steps:

1. Data Collection

- Download dataset from Kaggle

2. Data Preprocessing

- Handle missing values
- Remove duplicate records

- Convert data types if necessary

3. Exploratory Data Analysis (EDA)

- Analyze booking trends
- Study cancellation patterns
- Identify relationships between variables

4. Data Visualization

- Use charts such as:
    - Bar charts
    - Line plots
    - Pie charts
    - Distribution plots

5. Tools & Technologies

- **Python :** Python is a high-level programming language used for data analysis, machine learning, and software development. It is easy to learn and widely used for handling and analyzing data.
- **Pandas :** Pandas is a Python library used for data manipulation and analysis. It helps in working with datasets, especially in tabular form (rows and columns), and provides functions for cleaning, filtering, and processing data.
- **Matplotlib :** Matplotlib is a Python library used for creating basic graphs and charts such as bar charts, line plots, and histograms. It helps in visualizing data in a simple way.
- **Seaborn :** Seaborn is a data visualization library built on top of Matplotlib. It provides more attractive and advanced graphs like heatmaps, distribution plots, and statistical charts.
- **Jupyter Notebook :** Jupyter Notebook is an interactive environment where you can write and run Python code, add explanations, and display graphs in one place. It is widely used for data analysis and reporting.

---

7. Expected Outcome

- Understanding of hotel booking patterns

- Identification of major reasons for cancellations
- Insights into seasonal trends
- Visual representation of important data patterns

---

8. Conclusion

This project will help in understanding customer behavior in the hotel industry using data analysis techniques. The insights gained can help improve decision-making and reduce booking cancellations.